Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

B. E.(COE/CSE/EE) : MST **Course Code: UCS510/UCS319/ULC303**

Course Name: Computer Architecture and Organization **Date & Time 01/July/2026 8:00 AM**

Duration: 2 Hours, **M. Marks: 30**

Faculty: Dr. Abdul Kadir

Note: All questions carry equal marks. Attempt all questions in order.

Q.No.	Question	Marks	CO	BL
Q1.A	With reference to multiplexers as combinational logic building blocks, answer the following: a. Draw and describe the operation of a 4-to-1 multiplexer. b. Design a 32-to-1 line multiplexer using four 8-to-1 line multiplexers. c. How can a multiplexer be used to implement a Boolean function? Give an example.	6	CO1	BL3
Q01.B	The content of the PC in the basic computer is 5C8 (all numbers are in hexadecimal). The content of AC is 3DA7. The content of memory at address 5C8 is 716F. The content of memory at address 16F is 0E92. The content of memory at address E92 is 4A6B. a. What is the instruction that will be fetched and executed next? b. Show the binary operation that will be performed in the AC when the instruction is executed.	4	CO1	BL4
Q02.A	Consider three registers R1, R2, and R3 that store numbers in IEEE-754 single precision floating point format. Assume that R1 and R2 contain the values (in hexadecimal notation) 0x41200000 and 0x40A00000, respectively. If R3 = R1 + R2, determine the value stored in R3 in hexadecimal notation. Show all your work.	4	CO3	BL3
Q02.B	Simplify the following Boolean functions using four-variable maps. For each function, also determine the number of Prime Implicants (PIs) and Essential Prime Implicants (EPIs). a. $F(A, B, C, D) = \Sigma(1, 3, 5, 9)$ b. $F(A, B, C, D) = \Sigma(2, 6, 9, 10, 13, 14)$ c. $F(A, B, C, D) = \Sigma(0, 2, 3, 6, 7, 9, 13, 15)$	6	CO1	BL4
Q03.A	A computer uses a memory unit with 512K words of 36 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, a register code part to specify one of 64 registers, and an address part. a. How many bits are there in the operation code, the register code part, and the address part? b. Draw the instruction word format and indicate the number of bits in each part.	4	CO2	BL3

<table>
<tr><td></td><td>c. How many bits are there in the data and address inputs of the memory?</td><td></td><td></td><td></td></tr>
<tr><td>Q03.B</td><td>Consider the instruction formats and reference instruction set of a basic computer. For each of the following 16-bit instructions, give the equivalent four-digit hexadecimal code and explain in your own words what it is that the instruction is going to perform.

a. 0001 0011 0100 1010
b. 1010 0100 0101 1100
c. 0101 0001 0010 0000</td><td>3</td><td>CO2</td><td>BL2</td></tr>
<tr><td>Q03.C</td><td>Consider an instruction ADD M, R1 where:
i. Operand 1 (R1) uses Register Indirect addressing mode.
ii. Operand 2 (M) uses Memory Indirect addressing mode, with the instruction's address field specifying memory location 900.

Given the following initial contents:

<table><tr><th>Register/Memory</th><th>Content</th></tr><tr><td>R1</td><td>500</td></tr><tr><td>Memory[500]</td><td>1200</td></tr><tr><td>Memory[900]</td><td>600</td></tr><tr><td>Memory[600]</td><td>350</td></tr></table>
For each operand, determine:
a. The effective address of the operand
b. The number of register accesses and memory accesses required to fetch each operand.
c. The final value of each operand fetched</td><td>3</td><td>CO2</td><td>BL4</td></tr>
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